

Question	Answer	Marks
6(a)(i)	conversion (from a.c.) to d.c.	B1
6(a)(ii)	half-wave: voltage in one direction is removed	B1
	full-wave: voltage in one direction is reversed	B1
6(b)(i)	one gap connected by a single diode and other gap connected directly	B1
	diode drawn (in a circuit) with correct circuit symbol	B1
6(b)(ii)	smoothing	B1
6(c)(i)	$E = \frac{1}{2}CV^2$	C1
	$C = 2 \times 0.041 / 12^2 = 5.7 \times 10^{-4} \text{ F} = 570 \mu\text{F}$	A1
6(c)(ii)	$8.0 = 12.0 \exp(-0.010 / RC)$	C1
	$\ln(8.0 / 12.0) = -0.010 / (R \times 5.7 \times 10^{-4})$	C1
	$R = 43 \Omega$	A1

9702/41

Cambridge International AS & A Level – Mark Scheme

October/November 2024

PUBLISHED

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